ReadmitGuard: End-to-End Readmission Analysis

Integrated final report in PDF format covering Excel, SQL, Python, and Tableau

This report combines the full project workflow from spreadsheet exploration through SQL querying, Python EDA, and Tableau dashboarding. It answers the assignment questions in sequence, shows direct copies of the main code used in SQL and Python, summarizes the Excel workbook tabs, and connects the findings across tools to support practical recommendations for reducing 30-day hospital readmissions.

• Dataset: diabetic_data.csv with 101,766 patient encounters

• Core business question: Which patient characteristics and utilization patterns are most associated with readmission?

• Final interpretation across tools: readmission risk rises with age, clinical complexity, and prior utilization

Executive Summary

Across every tool, the same pattern appeared: patients with higher treatment intensity and older age profiles are more likely to be readmitted. Excel quickly established the basic distributions, SQL quantified the major encounter patterns and readmission rates, Python exposed the structure of the data and the relationships among key numerical variables, and Tableau translated those results into a dashboard that made the risk clusters easy to see. The most consistent signals were age, medication burden, lab volume, diagnosis count, and prior inpatient or emergency utilization.

1. Excel Findings

Question 1: Make a statistical summary of pertinent features like the number of lab procedures, procedures, drugs, and furthermore.

Metric	num_lab_procedures	num_procedures	num_medications
count	101766.0	101766.0	101766.0
mean	43.09564098028811	1.339730361810428	16.02184423088261
std	19.67436224914212	1.705806979121159	8.127566209167284
min	1.0	0.0	1.0
25%	31.0	0.0	10.0
50%	44.0	1.0	15.0
75%	57.0	2.0	20.0
max	132.0	6.0	81.0

Interpretation: The average encounter includes 43.10 lab procedures, 1.34 procedures, and 16.02 medications. The spread is wide, especially for medications, indicating strong variation in clinical complexity across patients.

Question 2: Create a report showing the number of readmissions by gender and age group.

gender	age	<30	>30	NO
Female	[0-10)	1	13	69
Female	[10-20)	24	147	231

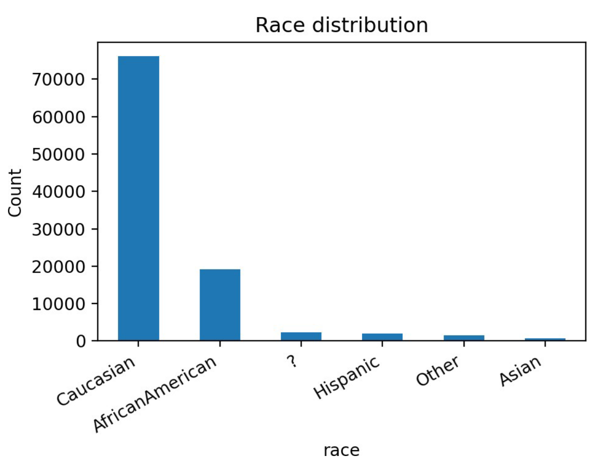
gender	age	<30	>30	NO
Female	[20-30)	177	346	591
Female	[30-40)	242	647	1273
Female	[40-50)	511	1685	2615
Female	[50-60)	851	3105	4616
Female	[60-70)	1206	4012	5843
Female	[70-80)	1635	5108	7242
Female	[80-90)	1282	3855	5378
Female	[90-100)	223	600	1180

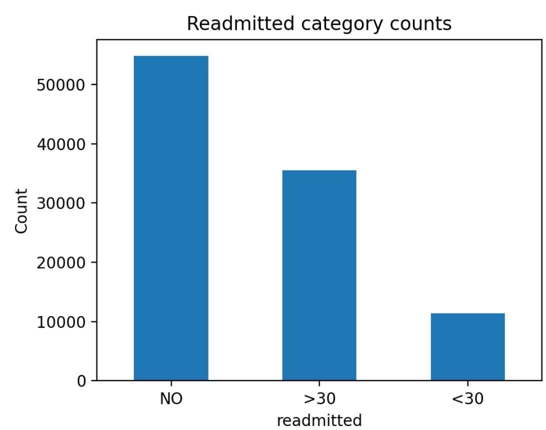
gender	age	<30	>30	NO
Male	[0-10)	2	13	63
Male	[10-20)	16	77	196
Male	[20-30)	59	164	320
Male	[30-40)	182	540	891
Male	[40-50)	516	1593	2765
Male	[50-60)	817	2812	5055
Male	[60-70)	1296	3885	6240
Male	[70-80)	1434	4367	6280
Male	[80-90)	796	2368	3518
Male	[90-100)	87	208	495

gender	age	<30	>30	NO
Unknown/Invalid	[60-70)	0	0	1
Unknown/Invalid	[70-80)	0	0	2

Interpretation: Readmissions rise sharply in the older age groups, especially 50-60, 60-70, and 70-80. Female and male patterns are broadly similar, with larger absolute volumes in the same older brackets.

Questions 3 and 4: Show race distribution in a chart; plot count of patients for each category of readmitted.





Race	Count
Caucasian	76099
AfricanAmerican	19210
?	2273
Hispanic	2037
Other	1506
Asian	641

Readmitted	Count
NO	54864
>30	35545
<30	11357

Interpretation: Caucasian encounters dominate the dataset, followed by African American patients. The readmission distribution shows 54,864 encounters with no readmission, 35,545 with readmission after 30 days, and 11,357 within 30 days. Even though the under-30 group is smaller, it remains the most operationally important outcome because it is the direct target for prevention.

2. SQL Findings

This section shows the assignment-ready SQL workflow and the results generated from the same dataset. The code is in the appendix.

Total Encounters
101766

Interpretation: The table contains 101,766 encounters, which sets the denominator for every later rate and proportion.

Top 10 most frequent diagnoses

Diagnosis	Frequency
428	6862
414	6581
786	4016
410	3614
486	3508
427	2766
491	2275
715	2151
682	2042
434	2028

Interpretation: A small set of diagnosis codes accounts for a large share of encounters. This concentration implies that the readmission problem is being driven by recurring disease profiles rather than a completely diffuse set of cases.

Average length of hospital stay for each admission type

Admission Type ID	Avg Length of Stay
7	4.86
2	4.61
6	4.58
1	4.38
3	4.32
5	3.95
4	3.2
8	3.06

Interpretation: Average stay varies by admission type, suggesting that the intake pathway matters and may proxy for severity or resource intensity.

Determine the number of readmitted patients and the percentage of total encounters that they represent

Total Encounters	Readmitted Count	Readmitted Percentage
101766	46902	46.09

Interpretation: 46,902 encounters eventually resulted in a readmission, or 46.09 percent of all encounters. That makes readmission prevention a meaningful operational target rather than a fringe issue.

Identify the age distribution of patients

Age Group	Patient Count
[0-10)	161

[10-20)	691
[20-30)	1657
[30-40)	3775
[40-50)	9685
[50-60)	17256
[60-70)	22483
[70-80)	26068
[80-90)	17197
[90-100)	2793

Interpretation: The dataset is concentrated in older age groups, especially 50-80. That pattern appears again in Excel, Python, and Tableau, which strengthens confidence that age is a genuine risk signal.

Identify the most common procedures performed during patient encounters

Num Procedures	Frequency
0	46652
1	20742
2	12717
3	9443
6	4954
4	4180
5	3078

Interpretation: Most encounters involve zero to two procedures. A smaller subset requires more intensive intervention, which is the group more likely to overlap with higher medication and lab counts.

Calculate the average number of medications prescribed for patients in each age group

Age Group	Avg Medications
[0-10)	6.18
[10-20)	8.28
[20-30)	11.97
[30-40)	14.09
[40-50)	15.39
[50-60)	16.58
[60-70)	17.15
[70-80)	16.41
[80-90)	15.33
[90-100)	13.82

Interpretation: Medication burden rises with age up to the 60-70 band and stays high through the older brackets. This directly connects to the complexity pattern seen in the Tableau bubble chart.

Identify the distribution of readmission rates across different payer codes

Interpretation: Readmission rates vary across payer codes. Smaller payer groups can show high percentages, while larger groups contribute more total readmission volume. That means both rate and scale matter for intervention planning.

payer_code	Total Encounters	Readmitted <30	Readmit 30-Day Rate %
OG	1033	136	13.17
SI	55	7	12.73
MD	3532	416	11.78
MC	32439	3810	11.75

DM	549	64	11.66
?	40256	4627	11.49
MP	79	9	11.39
HM	6274	644	10.26
CM	1937	198	10.22
SP	5007	510	10.19

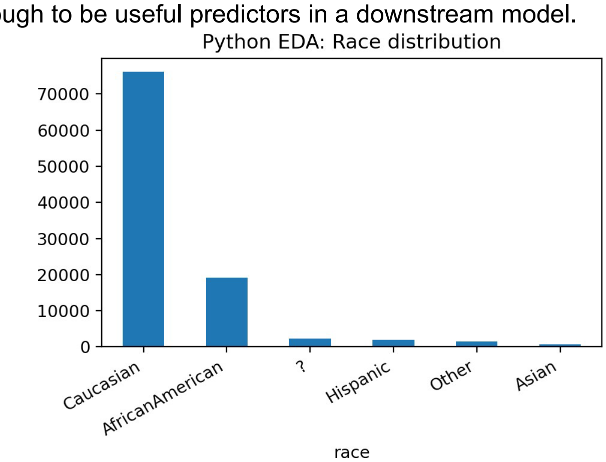
3. Python Exploratory Data Analysis (EDA) Findings

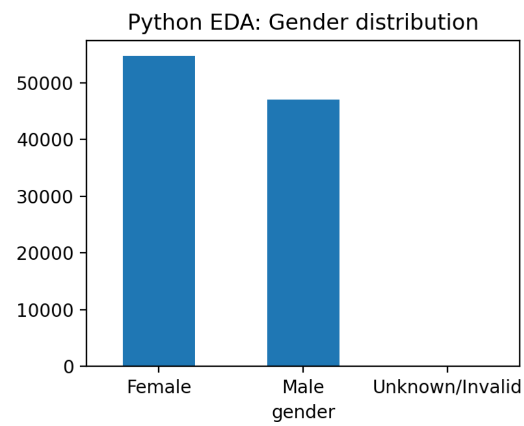
The Python work expanded the spreadsheet and SQL findings by profiling distributions, relationships, and outliers. The code is in the appendix.

Descriptive statistical analysis for numerical features

index	time_in_hospital	num_lab_procedures	num_procedures	num_medications	number_emergency	number_inpatient	number_diagnoses
count	101766.0	101766.0	101766.0	101766.0	101766.0	101766.0	101766.0
mean	4.4	43.1	1.34	16.02	0.2	0.64	7.42
std	2.99	19.67	1.71	8.13	0.93	1.26	1.93
min	1.0	1.0	0.0	1.0	0.0	0.0	1.0
25%	2.0	31.0	0.0	10.0	0.0	0.0	6.0
50%	4.0	44.0	1.0	15.0	0.0	0.0	8.0
75%	6.0	57.0	2.0	20.0	0.0	1.0	9.0
max	14.0	132.0	6.0	81.0	76.0	21.0	16.0

Interpretation: Numerical features show substantial spread. Medication count, lab procedures, and inpatient history all vary enough to be useful predictors in a downstream model.

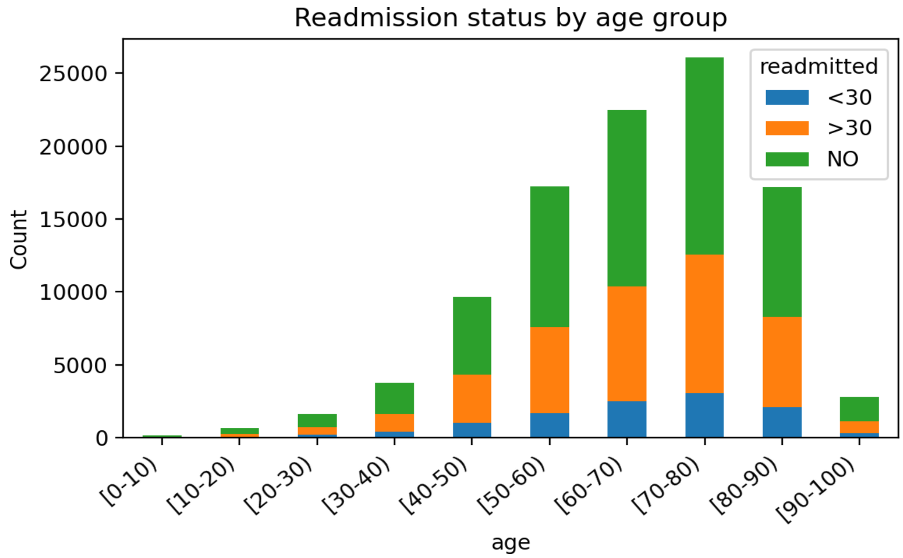




Interpretation: Race and gender are unevenly distributed, which matters for both modeling and fairness review. Any predictive model should be checked for disproportionate performance across demographic segments.

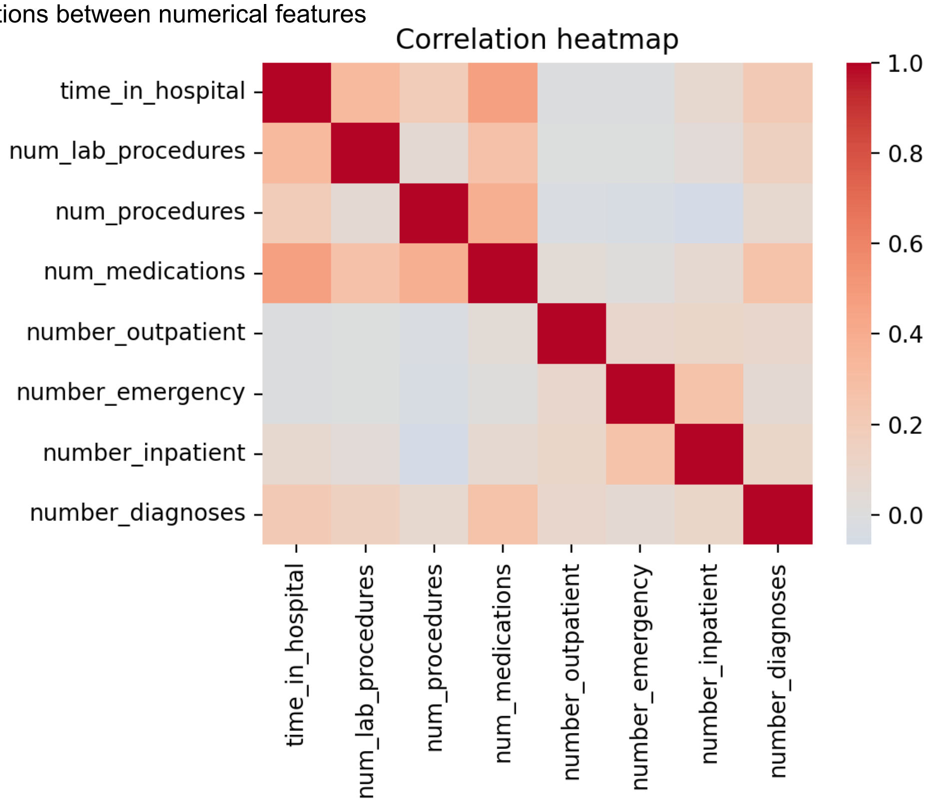
Explore the relationship between readmission status and age

age	<30	>30	NO
[0-10)	3	26	132
[10-20)	40	224	427
[20-30)	236	510	911
[30-40)	424	1187	2164
[40-50)	1027	3278	5380
[50-60)	1668	5917	9671
[60-70)	2502	7897	12084
[70-80)	3069	9475	13524
[80-90)	2078	6223	8896
[90-100)	310	808	1675



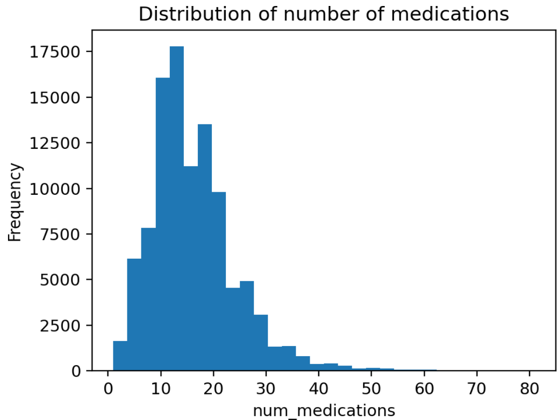
Interpretation: Readmission counts increase steeply in the older age buckets, especially 60-80. This pattern matches the Excel pivot table and the Tableau age view.

Investigate correlations between numerical features



Interpretation: Time in hospital is moderately correlated with num_medications (0.47) and num_lab_procedures (0.32). Num_procedures also correlates with num_medications (0.39). These are exactly the kinds of relationships expected in more complex, higher-touch encounters.

Python, continued



Medication Change	Count
No	54755
Ch	47011

Analyze the distribution of medication changes and total medications taken: The histogram is right-skewed, showing that most encounters involve a moderate number of medications but a smaller subset carries very high medication burden. The change field is also unevenly split, which signals that medication changes may be analytically useful.

Diagnosis Code	Count
428	6862
414	6581
786	4016
410	3614
486	3508
427	2766
491	2275
715	2151
682	2042
434	2028

Admission Type ID	Count
1	53990
3	18869
2	18480
6	5291
5	4785
8	320
7	21
4	10

Admission Source ID	Count
7	57494
1	29565
17	6781
4	3187
6	2264
2	1104
5	855
3	187
20	161
9	125

Discharge Disposition ID	Count
1	60234
3	13954
6	12902
18	3691
2	2128
22	1993
11	1642
5	1184
25	989
4	815

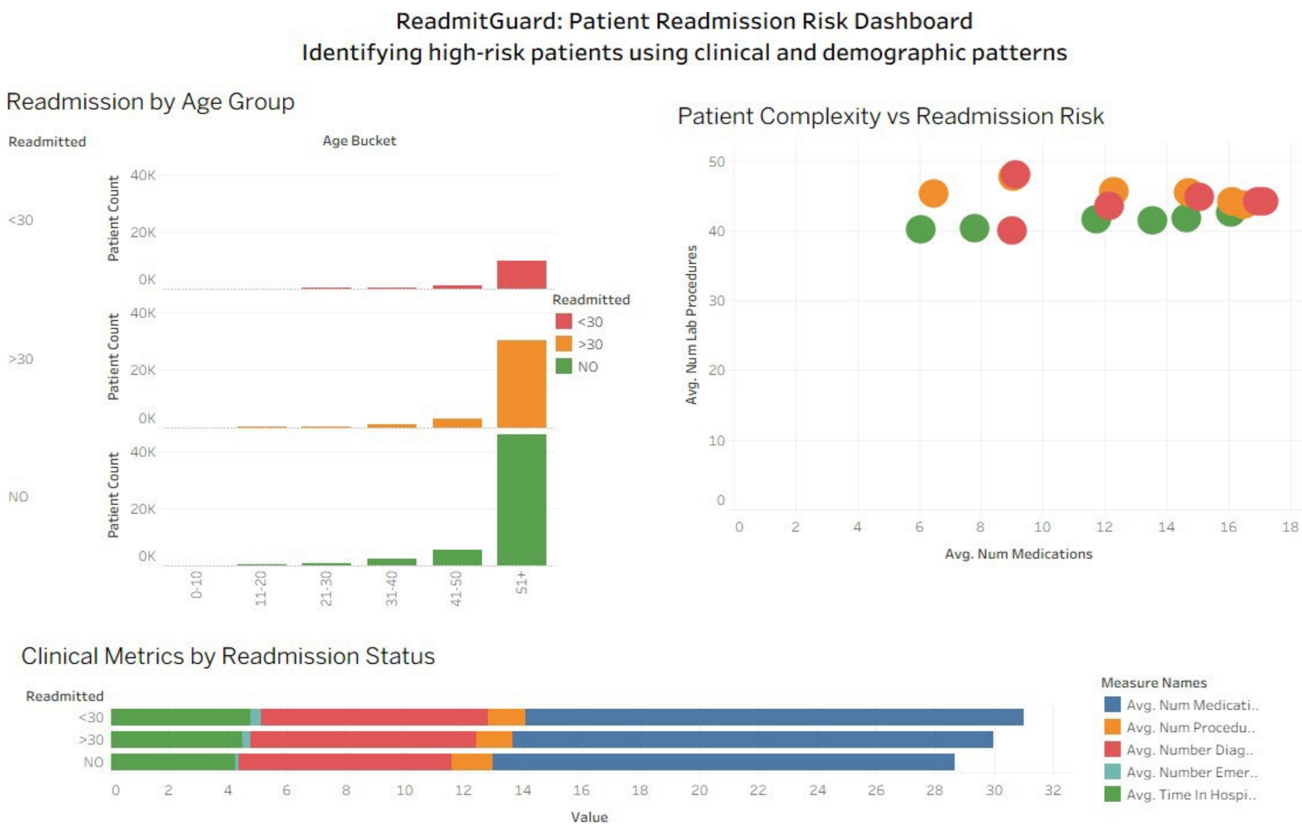
Interpretation: Diagnosis categories are concentrated in a short list of codes, while admission source and discharge disposition distributions show that a few pathways dominate the dataset. This means targeted operational changes could influence a large share of encounters.

Outlier review and fraud-detection framing

In a fraud detection framing for healthcare management, the same EDA techniques serve a different purpose. Instead of using extreme values only as clinical risk markers, analysts would review them as potential billing, documentation, or utilization anomalies. Encounters with unusually high medication counts, repeat emergency visits, or outlier combinations of labs, diagnoses, and procedures could indicate either very complex patients or records that deserve audit review. That distinction is why EDA is essential: it helps separate expected clinical variation from suspicious patterns before a formal model is built.

4. Tableau Dashboard Findings

The Tableau dashboard pulled the major SQL and Python results into an interactive visual summary. The screenshot below is the final dashboard view used for this report.



https://public.tableau.com/app/profile/joanne.perley/viz/HealthDashboard_17768223240600/Dashboard1

• The age chart shows that patient counts and readmission counts both concentrate heavily in the 51+ bucket. This visually confirms the age pattern already found in Excel and SQL.

• The bubble chart shows that higher average medications, diagnoses, and lab procedures cluster with readmitted categories, reinforcing the idea that clinical complexity is one of the strongest drivers of risk.

• The multi-metric comparison shows that readmitted groups consistently carry higher average clinical burden than the non-readmitted group across multiple measures.

5. Connections Across Excel, SQL, Python, and Tableau

• Excel established the broad shape of the problem. It showed that readmissions are concentrated in older groups and that the under-30 category, while smaller, is still substantial enough to matter operationally.

• SQL quantified the same pattern. It calculated 46.09 percent overall readmissions and showed that older age bands and high-medication groups are common in the dataset.

• Python explained why those patterns matter analytically. The descriptive statistics and correlations showed that time in hospital, medications, procedures, labs, and diagnoses all move together in clinically meaningful ways.

• Tableau made the findings visible in one place. The dashboard connected age, readmission status, and clinical complexity in a format that a nontechnical decision-maker could interpret quickly.

6. Suggestions Moving Forward

• Build a 30-day readmission risk score using age, num_medications, num_lab_procedures, number_diagnoses, number_inpatient, number_emergency, and time_in_hospital as core predictors.

• Prioritize discharge planning for older adults and for patients with high medication burden or repeated utilization history.

• Use the payer-code analysis to separate high-rate groups from high-volume groups, because those require different operational responses.

• Introduce routine fairness checks before deploying any predictive model, especially by race, gender, and age bucket.

• Adapt the Python outlier checks into a second workflow for anomaly detection so the same infrastructure can support both readmission prevention and potential fraud or audit review.

7. Conclusion

Taken together, the tools tell one consistent story: readmission is not random. It is concentrated among older patients and among encounters with heavier clinical burden. That makes the problem actionable. ReadmitGuard can use these patterns to target post-discharge support, improve care coordination, and eventually deploy predictive and anomaly-detection models that reduce avoidable readmissions while strengthening oversight.

Appendix A. Direct Copy of SQL Code Used

```
SELECT COUNT(*) AS total_encounters
FROM diabetic_data;

SELECT diag_1 AS diagnosis,
       COUNT(*) AS frequency
FROM diabetic_data
GROUP BY diag_1
ORDER BY frequency DESC
LIMIT 10;

SELECT admission_type_id,
       AVG(time_in_hospital) AS avg_length_of_stay
FROM diabetic_data
GROUP BY admission_type_id
ORDER BY avg_length_of_stay DESC;

SELECT COUNT(*) AS total_encounters,
       SUM(CASE WHEN readmitted IN ('<30','>30') THEN 1 ELSE 0 END) AS readmitted_count,
       ROUND(SUM(CASE WHEN readmitted IN ('<30','>30') THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS readmitted_percentage
FROM diabetic_data;

SELECT age,
       COUNT(*) AS patient_count
FROM diabetic_data
GROUP BY age
ORDER BY age;

SELECT num_procedures,
       COUNT(*) AS frequency
FROM diabetic_data
GROUP BY num_procedures
ORDER BY frequency DESC;

SELECT age,
       AVG(num_medications) AS avg_medications
FROM diabetic_data
GROUP BY age
ORDER BY age;

SELECT payer_code,
       COUNT(*) AS total_encounters,
       SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) AS readmit_30,
       ROUND(SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS readmit_30_rate
FROM diabetic_data
GROUP BY payer_code
ORDER BY readmit_30_rate DESC;
```

Appendix B. Direct Copy of Python EDA Code Used

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('diabetic_data.csv')

# 1. Descriptive statistical analysis for numerical features
desc = df.describe()

# 2. Visualize categorical distributions
df['race'].value_counts().plot(kind='bar')
df['gender'].value_counts().plot(kind='bar')

# 3. Relationship between readmission status and age
readmission_by_age = pd.crosstab(df['age'], df['readmitted'])

# 4. Correlations between numerical features
corr = df.select_dtypes(include='number').corr()

# 5. Distribution of medication changes and total medications taken
change_counts = df['change'].value_counts()
plt.hist(df['num_medications'], bins=30)

# 6. Distribution of diagnosis categories
diag_counts = df['diag_1'].value_counts().head(10)

# 7. Distribution across admission types, sources, and discharge dispositions
admission_counts = df['admission_type_id'].value_counts()
source_counts = df['admission_source_id'].value_counts()
discharge_counts = df['discharge_disposition_id'].value_counts()

# 8. Outliers in numerical features
plt.boxplot(df['num_medications'])
plt.boxplot(df['num_lab_procedures'])
```